Systems of Linear Equations

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Part I: Gaussian Elimination Method

General Theory

A system of linear equations

$$AX = B$$

can be solved using Gaussian elimination by reducing the augmented matrix $[A|B]$ to row echelon form using elementary row operations. The solution is then obtained by back substitution.

Consider a system of m linear equations in n unknowns

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\
\vdots \qquad\qquad\qquad &\quad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m.
\end{aligned}$$

This system can be written in matrix form as

$$AX = B,$$

where

$$A = [a_{ij}]_{m\times n}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

The Gaussian elimination method is a systematic procedure for solving the system by transforming it into an equivalent system whose coefficient matrix is in row echelon form.

Augmented Matrix

The augmented matrix of the system is defined as

$$[A|B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right].$$

Solving the system is equivalent to transforming $[A|B]$ into a simpler form using elementary row operations.

Elementary Row Operations

The following three operations are used in Gaussian elimination:

1. Interchange two rows:

$$R_i \leftrightarrow R_j.$$

2. Multiply a row by a non-zero scalar:

$$R_i \to cR_i, \quad c \neq 0.$$

3. Add a multiple of one row to another:

$$R_i \to R_i + cR_j.$$

These operations do not change the solution set of the system.

Row Echelon Form

A matrix is said to be in **row echelon form (REF)** if:

- All non-zero rows are above any rows of zeros.
- The leading non-zero entry (pivot) of each non-zero row is to the right of the pivot in the row above it.
- All entries below each pivot are zero.

Procedure of Gaussian Elimination

The Gaussian elimination method consists of the following steps:

1. Write the augmented matrix $[A|B]$.
2. Use elementary row operations to create zeros below the first pivot.
3. Continue the process to obtain row echelon form.
4. Apply back substitution to determine the values of the unknowns.

Back Substitution

After obtaining a row echelon form, the system has the form

$$\begin{aligned} u_{11}x_1 + u_{12}x_2 + \cdots + u_{1n}x_n &= c_1, \\ u_{22}x_2 + \cdots + u_{2n}x_n &= c_2, \\ \vdots \qquad\qquad\qquad &\ \ \vdots \\ u_{rr}x_r &= c_r, \end{aligned}$$

where r is the rank of A.

The unknowns are determined by solving the last equation first and then substituting successively into the preceding equations.

Nature of Solutions

Let n be the number of variables also called as unknowns. The nature of solutions is determined by comparing the ranks of A and $[A|B]$.

- If $\text{rank}(A) = \text{rank}([A|B]) = r = n$. The system has a unique solution.
- If $\text{rank}(A) = \text{rank}([A|B]) = r < n \implies (n-r)$-free variables. The system has infinitely many solutions.
- If $\text{rank}(A) \neq \text{rank}([A|B])$, the system has no solution.

Advantages

- It is systematic and easy to implement.
- It applies to any system of linear equations.
- It forms the basis of many numerical methods.

Problems on Gauss elimination method

Problem 1

$$\begin{aligned} x + y + z &= 6 \\ 2x + y + z &= 7 \\ x + 2y + z &= 8 \end{aligned}$$

Problem 2

$$
\begin{aligned}
2x + y - z &= 3 \\
4x + 2y - 2z &= 6 \\
x + y + z &= 5
\end{aligned}
$$

Problem 3

$$
\begin{aligned}
3x + 2y + z &= 1 \\
2x + y + 3z &= 4 \\
x + 4y + 2z &= 2
\end{aligned}
$$

Solution to Problem 3

*Step 1: Formation of Augmented Matrix The augmented matrix of the system is

$$
\left[\begin{array}{ccc|c}
3 & 2 & 1 & 1 \\
2 & 1 & 3 & 4 \\
1 & 4 & 2 & 2
\end{array}\right].
$$

*Step 2: First Elimination Interchange R_1 and R_3 to obtain a convenient pivot. $R_1 \leftrightarrow R_3$

$$
\left[\begin{array}{ccc|c}
1 & 4 & 2 & 2 \\
2 & 1 & 3 & 4 \\
3 & 2 & 1 & 1
\end{array}\right].
$$

Eliminate the entries below the first pivot. $R_2 \to R_2 - 2R_1, \quad R_3 \to R_3 - 3R_1$

$$
\left[\begin{array}{ccc|c}
1 & 4 & 2 & 2 \\
0 & -7 & -1 & 0 \\
0 & -10 & -5 & -5
\end{array}\right].
$$

*Step 3: Second Elimination Eliminate the entry below the second pivot. $R_3 \to R_3 - \frac{10}{7}R_2$

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 2 \\ 0 & -7 & -1 & 0 \\ 0 & 0 & -\frac{25}{7} & -5 \end{array}\right].$$

This is the row echelon form. *Step 4: Back Substitution From the third row,

$$-\frac{25}{7}z = -5 \quad \Rightarrow \quad z = \frac{7}{5}.$$

From the second row,

$$-7y - z = 0 \quad \Rightarrow \quad -7y = \frac{7}{5} \quad \Rightarrow \quad y = -\frac{1}{5}.$$

From the first row,

$$x + 4y + 2z = 2.$$

Substituting y and z,

$$x + 4\left(-\frac{1}{5}\right) + 2\left(\frac{7}{5}\right) = 2,$$

$$x - \frac{4}{5} + \frac{14}{5} = 2,$$
$$x + 2 = 2,$$
$$x = 0.$$

*Final Solution: Therefore, the solution of the system is

$$x = 0, \quad y = -\frac{1}{5}, \quad z = \frac{7}{5}.$$

$$\boxed{(x, y, z) = \left(0, -\frac{1}{5}, \frac{7}{5}\right)}.$$

Problem 4

$$\begin{aligned} 4x - y + 2z &= 7 \\ 3x + 2y - z &= 4 \\ 5x + y + z &= 9 \end{aligned}$$

Problem 5

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x + 4y + 6z &= 12 \\ 3x + 6y + 9z &= 18 \end{aligned}$$

Part II: Gauss–Jordan Elimination Method

General Theory

The Gauss–Jordan method reduces the augmented matrix $[A|B]$ directly to reduced row echelon form (RREF) using elementary row operations. The solution can then be read directly from the final matrix.

Consider a system of linear equations

$$AX = B,$$

where A is an $m \times n$ matrix, X is the vector of unknowns, and B is the constant vector.

The Gauss–Jordan elimination method is an extension of Gaussian elimination in which the augmented matrix $[A|B]$ is reduced completely to reduced row echelon form (RREF).

Reduced Row Echelon Form

A matrix is said to be in **reduced row echelon form** if:

- All zero rows are at the bottom.
- Each non-zero row has leading entry 1.
- Each leading 1 is the only non-zero entry in its column.
- The leading 1 of each row is to the right of the leading 1 in the previous row.

Procedure

1. Form the augmented matrix $[A|B]$.
2. Use elementary row operations to convert $[A|B]$ into RREF.
3. Read the solution directly from the final matrix.

Advantages

- No back substitution is required.
- Solutions are obtained directly.
- Useful for computing inverses.

Problems on Gauss–Jordan Elimination Method

Problem 6

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 11 \\ 2x + y + z &= 7 \end{aligned}$$

Problem 7

$$\begin{aligned} 2x + 3y - z &= 5 \\ 4x + 6y - 2z &= 10 \\ x + y + z &= 4 \end{aligned}$$

Problem 8

$$\begin{aligned} 3x + 2y + z &= 4 \\ 2x + 5y + 3z &= 7 \\ x + 3y + 4z &= 6 \end{aligned}$$

Solution to Problem 8

*Step 1: Write the Augmented Matrix

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 4 \\ 2 & 5 & 3 & 7 \\ 1 & 3 & 4 & 6 \end{array}\right]$$

*Step 2: Make Leading Entry of Row 1 Equal to 1 Interchange R_1 and R_3:

$$R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 6 \\ 2 & 5 & 3 & 7 \\ 3 & 2 & 1 & 4 \end{array}\right]$$

*Step 3: Eliminate Entries Below First Pivot

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 6 \\ 0 & -1 & -5 & -5 \\ 0 & -7 & -11 & -14 \end{array}\right]$$

*Step 4: Make Second Pivot Equal to 1

$$R_2 \rightarrow -R_2$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 6 \\ 0 & 1 & 5 & 5 \\ 0 & -7 & -11 & -14 \end{array}\right]$$

*Step 5: Eliminate Other Entries in Column 2

$$R_1 \rightarrow R_1 - 3R_2$$

$$R_3 \rightarrow R_3 + 7R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -11 & -9 \\ 0 & 1 & 5 & 5 \\ 0 & 0 & 24 & 21 \end{array}\right]$$

*Step 6: Make Third Pivot Equal to 1

$$R_3 \rightarrow \frac{1}{24}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -11 & -9 \\ 0 & 1 & 5 & 5 \\ 0 & 0 & 1 & \frac{7}{8} \end{array}\right]$$

*Step 7: Eliminate Other Entries in Column 3

$$R_1 \to R_1 + 11R_3$$

$$R_2 \to R_2 - 5R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{5}{8} \\ 0 & 1 & 0 & \frac{5}{8} \\ 0 & 0 & 1 & \frac{7}{8} \end{array}\right]$$

*Final Answer

$$\boxed{x = \frac{5}{8}, \quad y = \frac{5}{8}, \quad z = \frac{7}{8}}$$

Problem 9

$$\begin{aligned} x + y + z &= 3 \\ 2x + 2y + 2z &= 7 \\ 3x + 3y + 3z &= 9 \end{aligned}$$

Solution to Problem 9

*Step 1: Formation of Augmented Matrix

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 7 \\ 3 & 3 & 3 & 9 \end{array}\right].$$

*Step 2: First Elimination

Eliminate the entries below the first pivot.

$R_2 \to R_2 - 2R_1, \quad R_3 \to R_3 - 3R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

*Step 3: Reduced Row Echelon Form

The matrix is already in reduced row echelon form.

The second row represents the equation

$$0x + 0y + 0z = 1,$$

which is impossible.

*Step 4: Interpretation

Since the system leads to a contradiction, it is inconsistent.

Therefore, the system has no solution.

Problem 10

$$\begin{aligned} 2x + y + 3z &= 8 \\ 4x + 2y + 6z &= 16 \\ x + y + z &= 5 \end{aligned}$$

Part III: LU Decomposition Method

General Theory

The LU decomposition method factors the coefficient matrix as

$$A = LU,$$

where L is a lower triangular matrix with diagonal elements equal to 1 and U is an upper triangular matrix. The system is solved using forward and backward substitution.

Consider the system of linear equations

$$AX = B,$$

where A is a 3×3 coefficient matrix, X is the column vector of unknowns, and B is the constant vector.

The LU decomposition method factors A as

$$A = LU,$$

where:

- L is a lower triangular matrix with diagonal elements equal to 1,
- U is an upper triangular matrix.

The matrix U is obtained by reducing A to upper triangular form using **Gaussian elimination** (without row interchange).

The multipliers used during elimination are placed in the corresponding positions of L, while:

- diagonal entries of L are 1,
- upper triangular entries of L are 0.

That is, always assume the form of L to be:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix};$$

*Step 1: Factorization

Reduce A to an upper triangular matrix U using row operations of the form

$$R_i \to R_i - m_{ij}R_j.$$

Store the multipliers m_{ij} in the lower triangular matrix L. Notice that once you have $a'_{11} = 0$ and $a'_{21} = 0$ the corresponding multipliers are always

$$m_{21} = \frac{a_{21}}{a_{11}}, \quad \text{and} \quad m_{31} = \frac{a_{31}}{a_{11}}$$

and once you have $a''_{32} = 0$, the remaining multiplier is

$$m_{32} = \frac{a'_{32}}{a'_{22}}.$$

Then,

$$A = LU.$$

*Step 2: Forward Substitution Solve

$$LY = B$$

for Y using forward substitution.

*Step 3: Backward Substitution Solve

$$UX = Y$$

for X using backward substitution.
The vector X gives the solution of the system.

Problems on LU Decomposition Method

Problem 11

$$\begin{aligned} 2x + y + z &= 5 \\ 4x + 3y + 3z &= 11 \\ 2x + 2y + 3z &= 8 \end{aligned}$$

Solution to Problem 11 : LU Decomposition method

Consider the system

$$AX = B,$$

where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 2 & 2 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 11 \\ 8 \end{bmatrix}.$$

*Step 1: Reduction of A to Upper Triangular Form Apply Gaussian elimination. $R_2 \to R_2 - 2R_1, \quad R_3 \to R_3 - R_1$

$$A^{(1)} \sim \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Multipliers from A:

$$m_{21} = \frac{a_{21}}{a_{11}} = 2, \quad \text{and} \quad m_{31} = \frac{a_{31}}{a_{11}} = 1$$

Next,

$$R_3 \to R_3 - R_2$$

$$A^{(2)} \sim \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiplier from $A^{(1)}$:

$$m_{32} = \frac{a'_{32}}{a'_{22}} = 1$$

Hence,

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

*Step 2: Construction of L

Using the multipliers, we form

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Thus,

$$A = LU.$$

*Step 3: Forward Substitution ($LY = B$)

Let

$$Y = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}.$$

We solve

$$LY = B.$$

That is,

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \\ 8 \end{bmatrix}.$$

This gives

$$\alpha = 5,$$

$$2\alpha + \beta = 11,$$

$$\alpha + \beta + \gamma = 8.$$

Solving step by step,

$$\alpha = 5,$$

$$\beta = 11 - 2(5) = 1,$$

$$\gamma = 8 - 5 - 1 = 2.$$

Hence,

$$Y = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}.$$

*Step 4: Backward Substitution ($UX = Y$)

Now solve

$$UX = Y.$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}.$$

This yields

$$2x + y + z = 5,$$

$$y + z = 1,$$

$$z = 2.$$

$$y + 2 = 1 \quad \Rightarrow \quad y = -1.$$

From

$$2x - 1 + 2 = 5,$$

$$2x + 1 = 5,$$

$$x = 2.$$

Final Solution:

$$x = 2, \quad y = -1, \quad z = 2.$$

Problem 12

$$\begin{aligned} 3x + 2y + z &= 6 \\ 6x + 5y + 4z &= 15 \\ 3x + 3y + 2z &= 9 \end{aligned}$$

Problem 13

$$\begin{aligned} 4x + 2y + z &= 7 \\ 8x + 7y + 5z &= 20 \\ 4x + 5y + 4z &= 15 \end{aligned}$$

Problem 14

$$\begin{aligned} 5x + 2y + z &= 6 \\ 10x + 6y + 4z &= 15 \\ 5x + 4y + 3z &= 11 \end{aligned}$$

Problem 15

$$\begin{aligned} 6x + 2y + z &= 7 \\ 12x + 7y + 4z &= 20 \\ 6x + 5y + 6z &= 18 \end{aligned}$$

Part VI: Inverse of a Matrix Using Gauss–Jordan Method

General Theory

Let A be a square matrix of order 3.

The inverse of A, denoted by A^{-1}, exists if and only if

$$\det(A) \neq 0.$$

If A^{-1} exists, then

$$AA^{-1} = A^{-1}A = I,$$

where I is the identity matrix.

Gauss–Jordan Method

To find A^{-1} using Gauss–Jordan elimination, proceed as follows:

1. Form the augmented matrix
$$[A|I],$$
where I is the 3×3 identity matrix.
2. Apply elementary row operations to transform A into the identity matrix.
3. If A is reduced to I, then the augmented part becomes A^{-1}:
$$[I|A^{-1}].$$
4. If A cannot be reduced to I, then A is singular and has no inverse.

Remark

If during the process a row of zeros appears on the left side, then $\det(A) = 0$ and A^{-1} does not exist.

Problems on Finding Inverse Using Gauss–Jordan Method

Problem 16

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Problem 17

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Problem 18

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Problem 19

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution to Problem 19

We form the augmented matrix $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 4 & 1 & 2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 1 & 2 & 2 & 0 & 0 & 1 \end{array}\right]$$

Interchange R_1 and R_3:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 0 & 0 & 1 \\ 2 & 3 & 1 & 0 & 1 & 0 \\ 4 & 1 & 2 & 1 & 0 & 0 \end{array}\right]$$

Apply row operations:

$$R_2 \to R_2 - 2R_1, \quad R_3 \to R_3 - 4R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 0 & 0 & 1 \\ 0 & -1 & -3 & 0 & 1 & -2 \\ 0 & -7 & -6 & 1 & 0 & -4 \end{array}\right]$$

Multiply R_2 by -1:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 & -1 & 2 \\ 0 & -7 & -6 & 1 & 0 & -4 \end{array}\right]$$

Eliminate other entries in the second column:

$$R_1 \to R_1 - 2R_2, \quad R_3 \to R_3 + 7R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 0 & 2 & -3 \\ 0 & 1 & 3 & 0 & -1 & 2 \\ 0 & 0 & 15 & 1 & -7 & 10 \end{array}\right]$$

Divide R_3 by 15:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -4 & 0 & 2 & -3 \\ 0 & 1 & 3 & 0 & -1 & 2 \\ 0 & 0 & 1 & \frac{1}{15} & -\frac{7}{15} & \frac{2}{3} \end{array}\right]$$

Eliminate entries above the third pivot:

$$R_1 \to R_1 + 4R_3, \quad R_2 \to R_2 - 3R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{4}{15} & \frac{2}{15} & -\frac{1}{3} \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{2}{5} & 0 \\ 0 & 0 & 1 & \frac{1}{15} & -\frac{7}{15} & \frac{2}{3} \end{array}\right]$$

Hence, the inverse matrix is

$$A^{-1} = \begin{bmatrix} \frac{4}{15} & \frac{2}{15} & -\frac{1}{3} \\ -\frac{1}{5} & \frac{2}{5} & 0 \\ \frac{1}{15} & -\frac{7}{15} & \frac{2}{3} \end{bmatrix}.$$

Problem 20

$$A = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Part V: Gauss–Seidel Method

General Theory

The Gauss–Seidel method is an iterative technique for solving $AX = B$. New values are used immediately in each iteration. The method converges under suitable conditions such as diagonal dominance.

Consider a system of linear equations

$$AX = B,$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

The Gauss–Seidel method is an iterative technique used to obtain approximate solutions of $AX = B$.

It is especially effective when the coefficient matrix A is **diagonally dominant**.

*Step 1: Rearrangement of Equations

Rewrite each equation by solving for its leading variable:

$$x = \frac{1}{a_{11}} \left(b_1 - a_{12}y - a_{13}z \right),$$

$$y = \frac{1}{a_{22}} \left(b_2 - a_{21}x - a_{23}z \right),$$

$$z = \frac{1}{a_{33}} \left(b_3 - a_{31}x - a_{32}y \right).$$

*Step 2: Iterative Formula

Let $(x^{(k)}, y^{(k)}, z^{(k)})$ be the approximation at the k-th iteration.

Then the Gauss–Seidel formulas are:

$$x^{(k+1)} = \frac{1}{a_{11}} \left(b_1 - a_{12}y^{(k)} - a_{13}z^{(k)} \right),$$

$$y^{(k+1)} = \frac{1}{a_{22}} \left(b_2 - a_{21}x^{(k+1)} - a_{23}z^{(k)} \right),$$

$$z^{(k+1)} = \frac{1}{a_{33}} \left(b_3 - a_{31}x^{(k+1)} - a_{32}y^{(k+1)} \right).$$

Note that newly computed values are used immediately.

*Step 3: Initial Approximation

Choose an initial guess

$$(x^{(0)}, y^{(0)}, z^{(0)}),$$

usually $(0, 0, 0)$.

*Step 4: Convergence Criterion

Continue iterations until

$$|x^{(k+1)} - x^{(k)}|,\ |y^{(k+1)} - y^{(k)}|,\ |z^{(k+1)} - z^{(k)}|$$

are sufficiently small.

Remark on Convergence

The Gauss–Seidel method converges if A is strictly diagonally dominant, i.e.,

$$|a_{ii}| > \sum_{j\neq i} |a_{ij}|, \quad i = 1, 2, 3.$$

Problems on Gauss-Seidel Method

Problem 21

$$\begin{aligned} 10x - y + z &= 11 \\ -2x + 10y + z &= 10 \\ x + y + 10z &= 12 \end{aligned}$$

Solution to Problem 21: Gauss-Seidel Method

*Step 1: Rearrangement of Equations

Solving each equation for the leading variable, we obtain:

$$x = \frac{11 + y - z}{10},$$

$$y = \frac{10 + 2x - z}{10},$$

$$z = \frac{12 - x - y}{10}.$$

Hence, the Gauss–Seidel iteration formulas are:

$$x^{(k+1)} = \frac{11 + y^{(k)} - z^{(k)}}{10},$$

$$y^{(k+1)} = \frac{10 + 2x^{(k+1)} - z^{(k)}}{10},$$

$$z^{(k+1)} = \frac{12 - x^{(k+1)} - y^{(k+1)}}{10}.$$

*Step 2: Initial Approximation

Choose the initial approximation as

$$x^{(0)} = 0, \quad y^{(0)} = 0, \quad z^{(0)} = 0.$$

*Step 3: Iterations

*Iteration 1

$$x^{(1)} = \frac{11 + 0 - 0}{10} = 1.1,$$

$$y^{(1)} = \frac{10 + 2(1.1) - 0}{10} = 1.22,$$

$$z^{(1)} = \frac{12 - 1.1 - 1.22}{10} = 0.968.$$

$$(x^{(1)}, y^{(1)}, z^{(1)}) = (1.1,\ 1.22,\ 0.968).$$

*Iteration 2

$$x^{(2)} = \frac{11 + 1.22 - 0.968}{10} = 1.1252,$$

$$y^{(2)} = \frac{10 + 2(1.1252) - 0.968}{10} = 1.12824,$$

$$z^{(2)} = \frac{12 - 1.1252 - 1.12824}{10} = 0.974656.$$

$$(x^{(2)}, y^{(2)}, z^{(2)}) = (1.1252,\ 1.12824,\ 0.974656).$$

*Iteration 3

$$x^{(3)} = \frac{11 + 1.12824 - 0.974656}{10} = 1.11536,$$

$$y^{(3)} = \frac{10 + 2(1.11536) - 0.974656}{10} = 1.12561,$$

$$z^{(3)} = \frac{12 - 1.11536 - 1.12561}{10} = 0.975903.$$

$$(x^{(3)}, y^{(3)}, z^{(3)}) = (1.11536,\ 1.12561,\ 0.975903).$$

*Iteration 4

$$x^{(4)} = \frac{11 + 1.12561 - 0.975903}{10} = 1.11497,$$

$$y^{(4)} = \frac{10 + 2(1.11497) - 0.975903}{10} = 1.12540,$$

$$z^{(4)} = \frac{12 - 1.11497 - 1.12540}{10} = 0.97596.$$

$$(x^{(4)}, y^{(4)}, z^{(4)}) = (1.11497,\ 1.12540,\ 0.97596).$$

*Step 4: Convergence Check

Compare the results of Iterations 3 and 4:

$$|x^{(4)} - x^{(3)}| = 0.00039,$$

$$|y^{(4)} - y^{(3)}| = 0.00021,$$

$$|z^{(4)} - z^{(3)}| = 0.00006.$$

Since the differences are sufficiently small, the iterations have converged.

Final Approximate Solution

Therefore, the approximate solution of the system is

$$x \approx 1.115, \quad y \approx 1.125, \quad z \approx 0.976.$$

$$\boxed{(x, y, z) \approx (1.115,\ 1.125,\ 0.976)}.$$

Remark

The coefficient matrix is strictly diagonally dominant. Hence, the Gauss–Seidel method converges rapidly for this system.

Problem 22

$$\begin{aligned} 8x + y - z &= 8 \\ x + 7y + z &= 10 \\ -2x + y + 9z &= 9 \end{aligned}$$

Problem 23

$$\begin{aligned} x + 2y + 10z &= 12 \\ 10x + y + z &= 11 \\ 2x + 9y - z &= 10 \end{aligned}$$

Problem 24

$$\begin{aligned} 6x + 2y + z &= 9 \\ 2x + 5y + 2z &= 8 \\ x + 2y + 4z &= 7 \end{aligned}$$

Problem 25

$$\begin{aligned} 12x - 2y + z &= 11 \\ -3x + 11y + 2z &= 13 \\ 2x - y + 10z &= 10 \end{aligned}$$